

Implementation	Qubits	Modular Exponentiation		Circuit	
		Gates	Depth	Gates	Depth
Discrete Fourier transform	$2L$	$O(L^2)$	$O(L^2)$	$O(L^3)$	$O(L^3)$
Toffoli gates	$2L + 2$	$O(L^2 \log L)$	$O(L^2)$	$O(L^3 \log L)$	$O(L^3)$

Table 1: Circuit complexity for an L -bit integer $L = \lceil \log_2 N \rceil$